

Impact of Positional Encoding:

Clean and Adversarial Rademacher Complexity
for Transformers under In-Context Regression

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In-context learning (ICL): a Transformer solves a new task from a few examples in the prompt, with no weight updates.

Existing ICL theory usually analyzes a simplified attention model.

- The **positional encoding (PE)** is typically dropped, or absorbed into one uniform norm bound.
- Yet PE adds a whole matrix of trainable parameters.
- LLMs are also vulnerable to adversarial prompts (jailbreaks, prompt injection).

Question. How does a fully trainable PE affect

1. clean generalization, and
2. adversarial generalization when the prompt is attacked?



1. A Rademacher complexity bound for a one-layer Transformer under ICL regression that **decays with context length** t .
2. An explicit **PE cost**: trainable PE adds a complexity term that does not vanish as t grows.
3. An **adversarial** Rademacher bound: attacks enlarge the solution space, and PE adds a second, attack-dependent penalty.
4. Simulations that confirm each predicted trend.

PE widens the generalization gap, and **the widening is larger under attack.**



Setting: in-context linear regression

A prompt holds a context $S_t = \{(x_i, y_i)\}_{i=1}^t$ of t labeled examples and a query x_q whose label y_q is hidden:

$$X = \begin{pmatrix} x_1 & x_2 & \cdots & x_t & x_q \\ y_1 & y_2 & \cdots & y_t & 0 \end{pmatrix}^\top \in \mathbb{R}^{(t+1) \times (d+1)}.$$

The Transformer f_θ predicts $\hat{y}_q = f_\theta(X)$.

Data model (Assumption 4.1)

- $x_i \sim \mathcal{N}(0, I_d)$, $y_i = \mu^\top x_i$ (noiseless).
- $\mu \sim \mathcal{N}(0, I_d/d)$ is shared within a prompt and redrawn for each prompt.

Model: one layer, one head, trainable PE



$$H = XW_{\text{in}} + P$$

content embedding + positional encoding

$$H' = \sigma \left(\text{RowSoftmax} \left(\frac{HW_{QK}H^{\top}}{\sqrt{d_m}} \right) HW_V \right)$$

self-attention

$$f_{\theta}(X) = W_c^{\top} h'_q$$

linear readout at the query token

- $P \in \mathbb{R}^{(t+1) \times d_m}$ is **fully trainable**, with $\|P\|_F \leq B_P$ (d_m : embedding dimension).
- Trained with query-token MSE: $L(\theta) = \mathbb{E}(\hat{y}_q(S_t, x_q; \theta) - y_q)^2$.

Tool: Rademacher complexity



Proposition 3.1 (generalization bound)

Hypothesis class $\mathcal{H}$, loss $\ell \leq c$, sample S of m prompts; w.p. at least $1 - \delta$,

$$\sup_{h \in \mathcal{H}} \left(\underbrace{R(h)}_{\text{true risk}} - \underbrace{\widehat{R}_S(h)}_{\text{empirical risk}} \right) \leq 2 \text{Rad}_S(\ell \circ \mathcal{H}) + 4c \sqrt{\frac{2 \log(4/\delta)}{m}}.$$

Proof strategy

1. Dudley's integral reduces Rad_S to covering numbers.
2. Contraction peels off W_c, σ, W_V ; only (W_{in}, W_{QK}) remain.
3. Their covering number is controlled by the diameter of the **ICL solution space**.

Key object: the ICL solution space



Effective linear weight of the Transformer:

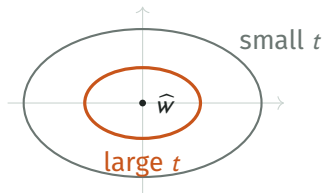
$$w_{\text{eff}} = \arg \min_w \mathbb{E}_{x_q} (\hat{y}_q - w^\top x_q)^2.$$

Model the realizable w_{eff} as a tolerance ellipsoid around the OLS solution $\hat{w}$:

$$\mathcal{W}_{S_t} = \{w : (w - \hat{w})^\top X_t^\top X_t (w - \hat{w}) \leq r\},$$

where $X_t = [x_1, \dots, x_t]^\top$ and $r > 0$ is a tolerance constant independent of t .

More examples shrink the ellipsoid: $\text{Diam}(\mathcal{W}_{S_t}) \lesssim O(\sqrt{r/t})$ for $t > d$.



Without PE: complexity decays with context length



The parameter space inherits this geometry, and Dudley's integral gives:

Theorem 4.1 (without PE)

$$\text{Rad}_S(\mathcal{F}_{S_t}) \lesssim O\left(\frac{L_f \sqrt{rD \log(t/r)}}{\sqrt{mt}}\right)$$

$\mathcal{F}_{S_t}$: Transformers constrained by context S_t ; L_f : Lipschitz constant of $\theta \mapsto f_\theta$;
 $D = d \cdot d_m + d_m^2$: number of parameters in (W_{in}, W_{QK}) .

- Decays like $\sqrt{\log t/t}$: more examples, better generalization.

With trainable PE: a complexity floor that does not decay



Write the PE model as the no-PE model plus a PE-induced bias:

$$f_{\text{PE}} = f_{\text{NoPE}} + f_{\text{bias}} \implies \text{Rad}_S(\mathcal{F}_{\text{PE}}) \leq \text{Rad}_S(\mathcal{F}_{\text{NoPE}}) + \text{Rad}_S(\mathcal{F}_{\text{bias}}).$$

Assumption 4.2: PE moves the effective weight by at most C_{PE} :

$$\|w_{\text{PE}} - w_{\text{NoPE}}\|_2 \leq C_{\text{PE}}.$$

Theorem 4.2 (with trainable PE)

$$\text{Rad}_S(\mathcal{F}_{S_t, \text{PE}}) \lesssim O\left(\frac{L_f \sqrt{rD \log(t/r)}}{\sqrt{mt}}\right) + \frac{C_{\text{PE}} \sqrt{d}}{\sqrt{m}}$$

The **PE term** does not decay with t , so it dominates for long contexts.



The attacker perturbs the **entire prompt** within a Frobenius ball of radius ε :

$$\tilde{\ell}_{\theta}(X, y_q) = \sup_{\|X' - X\|_F \leq \varepsilon} (\hat{y}_q(X'; \theta) - y_q)^2.$$

The inner sup is hard to analyze directly, so we use a surrogate loss.

Surrogate. If f_{θ} is L_x -Lipschitz in X ,

$$\tilde{\ell}_{\theta}(X, y) \leq \ell_{\text{rob}}(f_{\theta}(X), y) = (|f_{\theta}(X) - y| + L_x \varepsilon)^2.$$

ℓ_{rob} is L_h -Lipschitz in $f_{\theta}(X)$, so its complexity is at most L_h times that of the function class. The clean machinery then carries over to the *attacked* context S'_t .



Attack enlarges the solution space

Under attack, the solution space $\mathcal{W}'_{S_t}$ is the clean one scaled by Φ :

$$\text{Diam}(\mathcal{W}'_{S_t}) \lesssim O(\sqrt{r/t}) \cdot \Phi(\varepsilon, t, d), \quad \Phi(\varepsilon, t, d) = \frac{1}{1 - \sqrt{d/t} - \varepsilon/\sqrt{t}}, \quad \varepsilon < \sqrt{t} - \sqrt{d}.$$

- $\Phi > 1$ and increases with ε : the attack enlarges the complexity.
- Φ decreases with t : **longer prompts are more robust.**

Theorem 4.3 (adversarial, without PE)

$$\text{Rad}_{S'}(\tilde{\mathcal{L}}_{\mathcal{F}'_{S_t}}) \lesssim O\left(\frac{L_h L_f \sqrt{rD \log(t/r)}}{\sqrt{mt}}\right) \cdot \Phi(\varepsilon, t, d)$$

$\tilde{\mathcal{L}}_{\mathcal{F}'_{S_t}}$: class of adversarial losses $\tilde{\ell}_\theta$; S' : the attacked sample.

Main result: PE costs more under attack



Let $B = O\left(\frac{L_f \sqrt{rD \log(t/r)}}{\sqrt{mt}}\right)$ be the context term and $\Delta = \text{Rad}(\text{PE}) - \text{Rad}(\text{No PE})$.

Theorems 4.1–4.4 give:

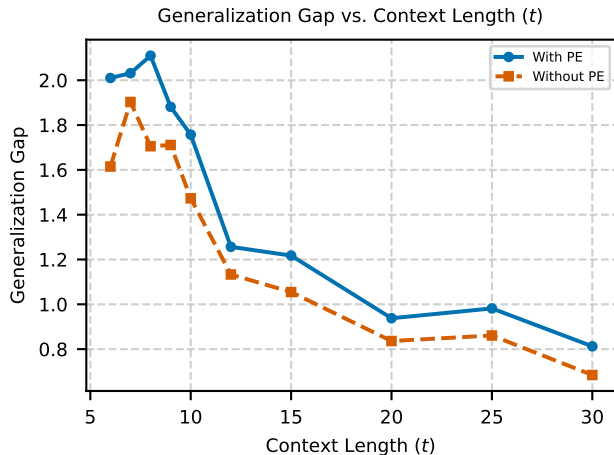
	No PE	Trainable PE	Gap Δ
Clean	B	$B + \frac{C_{\text{PE}} \sqrt{d}}{\sqrt{m}}$	$\frac{C_{\text{PE}} \sqrt{d}}{\sqrt{m}}$
Adversarial	$L_h B \Phi$	$L_h \left(B \Phi + \frac{C_{\text{PE}} \sqrt{d}}{\sqrt{m}} + \frac{C_{\text{PE}} \varepsilon}{\sqrt{m}} \right)$	$\approx \frac{C_{\text{PE}} \sqrt{d}}{\sqrt{m}} + \frac{C_{\text{PE}} \varepsilon}{\sqrt{m}}$

$\Delta_{\text{adv}} = \Delta_{\text{clean}} + O(\varepsilon/\sqrt{m})$: **the PE penalty grows with attack strength.**



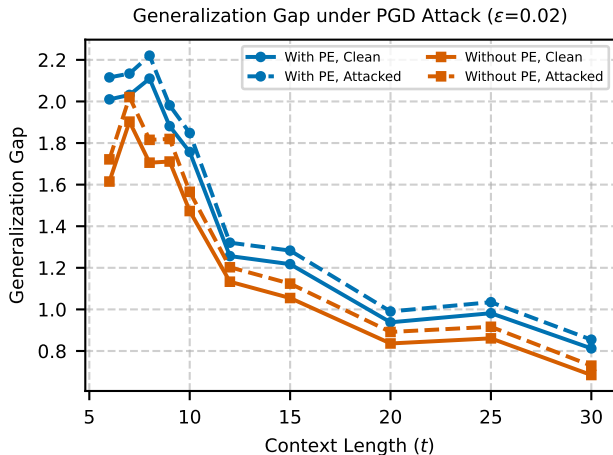
- **Data:** Assumption 4.1 with $d = 5$; 309 training tasks, 9,991 validation tasks.
- **Model:** one layer, one head, read-in layer, $d_m = 1024$ GPT2.
- **Training:** one fixed prompt of length t per training task; full-batch Adam, 3,000 epochs, so the empirical risk is measured cleanly.
- **Gap** = validation risk – training risk, on query-token MSE.
- **Attack:** PGD on the whole prompt, ℓ_2 budget ε , 40 steps of size $\varepsilon/10$.
- Mean over 6 seeds; context lengths $t > d$.

Trainable PE widens the clean gap



- The PE curve lies above the no-PE curve at every t (Thm 4.2).
- Both gaps decay with t (Thm 4.1).
- High variance near $t \approx d$: the interpolation threshold, as in double descent.

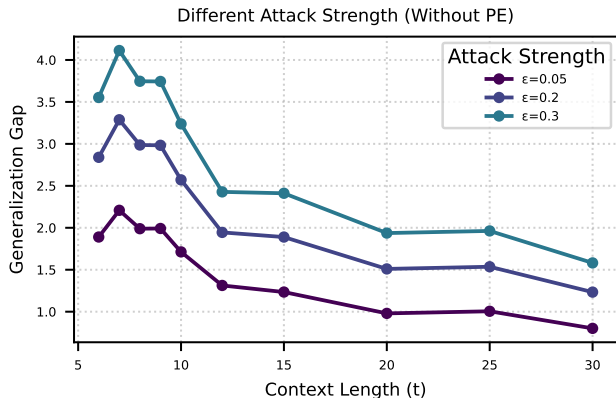
Under attack, the gap grows and PE stays worse



PGD with $\epsilon = 0.02$:

- attacked curves lie above clean curves (Thm 4.3);
- PE lies above no PE, clean or attacked (Thm 4.4);
- all gaps shrink as t grows.

Stronger attacks give larger gaps



No PE, varying ϵ :

- larger ϵ shifts the curve up;
- larger t lowers every curve;
- both match the behavior of $\Phi(\epsilon, t, d)$.



- 1. Longer context helps.** It shrinks the ICL solution space and tightens both clean and adversarial bounds.
- 2. Trainable PE has a cost.** It adds a complexity term $C_{PE}\sqrt{d}/\sqrt{m}$ that does not decay with context length.
- 3. Attacks magnify that cost.** The PE vs. no-PE gap gains an extra $O(\varepsilon/\sqrt{m})$ term.

Thank you